

LAB REPORT OCR SYSTEM - TECHNICAL DOCUMENTATION

Modular Architecture, YOLOv8 Integration, Fallback OCR Pipeline, and Data Model Design

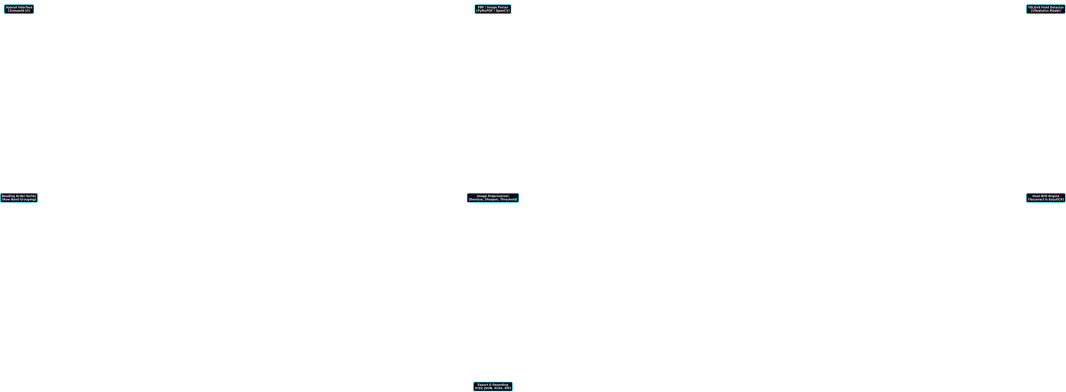
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1. Architecture Overview

The system architecture decomposes processing into independent modules: UI, Configuration, PDF parsing, YOLO Object Detection, OCR dispatcher, and Export helpers.



2. Module Specifications

Module Specifications Overview:

- app/app.py: Main Streamlit frontend interface layer.
- app/pipeline.py: Main processing workflow manager.
- app/detect.py: Bounding box reading order sorter.
- app/preprocess.py: Image filtering and enhancement filters.
- app/ocr.py: Multi-backend text parser runner.
- app/exports.py: CSV, JSON, and XLSX data serializer.
- app/pdf_utils.py: PDF parser rendering logic.

3. Pipeline Detail

The pipeline runner executes: YOLO inference -> reading order sort -> preprocessing -> crop -> OCR run -> clean output text -> build report structure data -> compile exports.